

Percentages

Turn every percentage into one multiplier. Forwards you multiply, backwards you divide, twice you multiply twice.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Percentages

01 — Percentage of an amount

145% of 1.80 m VAT at 20% commission, discount
is the target met?

02 — Percentage change and profit

% increase in population % profit which coupon saves more?

03 — Reverse percentages

sale price 20.40 after 15% off 15% more than March 17% of N is 357

04 — Repeated and combined changes

+15% then -8% "4 + 2.5 = 6.5" — is he right? $\times k$ each hour

05 — Percentages with fractions and ratio

mugs: $\frac{2}{5}$ and 32% bulbs 9:4:2 combined savings

THE MULTIPLIER Write the multiplier first — every question then becomes one calculation:

ZONE UP 12%

$\times 1.12$

100% + 12%

ZONE DOWN 15%

$\times 0.85$

100% - 15%

ZONE FORWARDS

old $\times m$

new value

ZONE BACKWARDS

new $\div m$

original value

01 Percentage of an Amount

Change the percentage to a decimal and multiply. Several parts? Work each one out, then add.

WHEN YOU SEE → YOU DO

“145% of 1.80 m”

→ 1.80×1.45

“plus VAT at 20%”

→ $\times 1.2$

“5% of the cost of all the car insurance he sells”

→ commission: $0.05 \times \text{total}$

“a target of 94%”

→ work out the actual %, then compare

“is it cheaper from ...”

→ work out both totals, then decide in words

WORKED MODEL • VAT AND PAYMENTS

A caravan costs £7000 plus VAT at 20%. Deposit £3000, the rest in 6 equal payments.

1 $7000 \times 1.2 = £8400$

add VAT

2 $8400 - 3000 = £5400$

after deposit

ANSWER $5400 \div 6 = £900$ a month

IDEA 01

Practise: Percentage of an Amount

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Multiplier

CH 1 • TOPIC 6 2 marks

A coat costs £85 plus VAT at 20%. Work out the total cost.

SELECTED PRACTICE**Q2** **PRACTICE** Practice • Easy

CH 1 • TOPIC 6 4 marks

2 Ria is going to buy a caravan.

The total cost of the caravan is £7000 **plus** VAT at 20%.

Ria pays a deposit of £3000 She pays the rest of the total cost in 6 equal monthly payments.

Work out the amount of each monthly payment.

(4 marks)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 6 1 mark**11** Work out 40 as a percentage of 10

Circle your answer.

4%	25%	300%	400%
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(1 mark)

Q4**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6 5 marks

4 Mr Watkins needs to buy some oil for his central heating.

Mr Watkins can put up to 1500 litres of oil in his oil tank.

There are already 850 litres of oil in the tank.

Mr Watkins is going to fill the tank with oil.

The price of oil is 67.2p per litre.

Mr Watkins gets 5% off the price of the oil.

How much does Mr Watkins pay for the oil he needs to buy?

(5 marks)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6 5 marks

8 Bill buys and sells laptops.

Last month Bill bought 50 laptops.

He paid £400 for each laptop.

He sold

40 of these laptops at a profit of 30% on each laptop

10 of these laptops at a profit of 15% on each laptop

Bill's target last month was to sell all 50 laptops for a total of at least £25 000

Did Bill reach this target?

(5 marks)

PAST PAPERS

Q6 **PAST PAPER** Edexcel 4MA1 • May 2024 • Paper 1HR • Q3**CH 1 • TOPIC 6** **5 marks**

- 3 Norberto sells white loaves of bread and brown loaves of bread.

He sells a total of 200 loaves such that

the number of white loaves sold : the number of brown loaves sold = 3 : 2

Norberto sells the white loaves for £1.50 each.

He sells the brown loaves for £1.75 each.

40% of the price of a white loaf is profit.

60% of the price of a brown loaf is profit.

Work out Norberto's total profit when he sells all 200 loaves.

£.....

(Total for Question 3 is 5 marks)

02 Percentage Change and Profit

Change over the **original**, times 100. Profit is the change from cost to selling price.

WHEN YOU SEE → YOU DO

“percentage increase / decrease”

$$\rightarrow \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

“percentage profit”

$$\rightarrow \frac{\text{selling} - \text{cost}}{\text{cost}} \times 100$$

“is Brigitte correct?” (two increases)

→ work out both %, compare

“which coupon saves more?”

→ work out each saving in money

Change = new – old

Divide by the ORIGINAL

× 100

ZONE %
change
original × 100

divide by where you **started**

ZONE PROFIT
total income – total cost

count **all** costs: delivery, materials, adhesive ...

WORKED MODEL • NOVEMBER 2021 P2H

Victor buys 12 bottles for \$21 and sells all 12 at \$2.45 each. Percentage profit?

1 $12 \times 2.45 = \$29.40$

income

2 $29.40 - 21 = \$8.40$

profit

ANSWER $\frac{8.40}{21} \times 100 = 40\%$

× **TRAP** Dividing by the new value: $\frac{8.40}{29.40}$ is wrong. The original (cost) always goes on the bottom.

IDEA 02

Practise: Percentage Change and Profit

My questions first, then selected practice, then real past papers.

Q7**DR ESLAM**

Percentage change

CH 1 • TOPIC 6

3 marks

A car's value fell from £12 500 to £9 800. Work out the percentage decrease.

SELECTED PRACTICE**Q8****PRACTICE**

Practice • Easy

CH 1 • TOPIC 6

3 marks

- 1 Bill's weight decreases from 64.8 kg to 59.3 kg.

Calculate the percentage decrease in Bill's weight.

Give your answer correct to 3 significant figures.

(3 marks)

Q9**PRACTICE**

Practice · Easy

CH 1 · TOPIC 6 3 marks

4 George wants to watch all 23 games that a football team will play at home next season.

He can buy

a season ticket costing £425

or 23 separate tickets costing £24 each ticket.

What percentage of the total cost of 23 separate tickets does George save by buying a season ticket?

(3 marks)**Q10****PRACTICE**

Practice · Easy

CH 1 · TOPIC 6 3 marks

13 In 1999 the minimum wage for adults was £3.60 per hour.

In 2013 it was £6.31 per hour.

Work out the percentage increase in the minimum wage.

(3 marks)

Q11**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6

4 marks

- 9 Renee buys 5 kg of sweets to sell.
She pays £10 for the sweets.

Renee puts all the sweets into bags.
She puts 250 g of sweets into each bag.
She sells each bag of sweets for 65p.

Renee sells all the bags of sweets.

Work out her percentage profit.

(4 marks)

PAST PAPERS

Q12**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q8

CH 1 • TOPIC 6 7 marks

- 8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	2016	2017	2018
Cost of insurance (euros)	500	545	592

Brigitte says,

“The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017”

- (a) Is Brigitte correct?

You must show how you get your answer.

Henri wants to insure his car.

He gets a discount of 15% off the normal price.

Henri pays 952 euros for his car insurance after the discount.

- (b) Work out the discount that Henri gets.

..... euros

(Total for Question 8 is 7 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q5

CH 1 • TOPIC 6 6 marks

- 5 In 2017, the population of a village was 7500
In 2019, the population of the village was 8265

(a) Work out the percentage increase in the population of the village from 2017 to 2019

..... %

In a sale, normal prices are reduced by 30%
The sale price of a T-shirt was 31.50 euros.

(b) Work out the normal price of the T-shirt.

..... euros

(Total for Question 5 is 6 marks)

Q14**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q3

CH 1 • TOPIC 6 6 marks

- 3 Gladys buys a table for \$465 to sell in her shop.

She sells the table for \$520

- (a) Work out the percentage profit that Gladys makes from the sale of the table.
Give your answer correct to 3 significant figures.

.....%

Gladys has a sale in her shop.

She decreases all the normal prices by 12%

The normal price of an armchair was \$550

- (b) Work out the sale price of the armchair.

\$.....

(Total for Question 3 is 6 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q6

CH 1 • TOPIC 6**3 marks**

- 6 Victor buys 12 bottles of apple juice for a total cost of \$21
Victor sells all 12 bottles at \$2.45 each bottle.

Work out Victor's percentage profit.

..... %

(Total for Question 6 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q4

CH 1 • TOPIC 6**5 marks**

- 4 Sarah makes and sells mugs.

One day she makes 150 mugs.

Her total cost for making these mugs is £1140

Of these mugs

$\frac{2}{5}$ are small mugs

32% are medium mugs

and the rest are large mugs

Here is Sarah's price list for selling each mug.

MUGS	
Small	£8.50
Medium	£11.20
Large	£14.20

Sarah sells all 150 mugs.

Work out her percentage profit.

Give your answer correct to the nearest whole number.

.....%

(Total for Question 4 is 5 marks)

Q17**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q8

CH 1 • TOPIC 6 5 marks

- 8 The table gives information about the average house price in England in 2018 and in 2019

Year	2017	2018	2019
Average house price (£)		228 314	231 776

- (a) Work out the percentage increase in the average house price from 2018 to 2019
Give your answer correct to one decimal place.

..... %

The average house price in 2019 was 7.7% greater than the average house price in 2017

- (b) Work out the average house price in 2017
Give your answer correct to 3 significant figures.

£

(Total for Question 8 is 5 marks)

Q18**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q2

CH 1 • TOPIC 6 7 marks

- 2 Mary saves for a holiday each year.

In 2020 she saved a total of \$720

In 2021, each month she saved \$78

The total amount Mary saved in 2021 was $P\%$ more than the total she saved in 2020

- (a) Work out the value of P

.....

Roberto is going to go on holiday.

He has two coupons that will save him money on his holiday.

Coupon A

18% off the cost of the
accommodation

Coupon B

12.5% off the total cost of the
accommodation **and** the flights

For Roberto's holiday

the cost of the accommodation is \$1600

the cost of the flights is \$800

Roberto can only use one of the coupons.

He wants to save as much money as he can.

- (b) Which of the two coupons, **A** or **B**, should he use?

Show your working clearly.

(Total for Question 2 is 7 marks)

Q19**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q6

CH 1 • TOPIC 6 7 marks

- 6 In his previous job, Pierre was paid 400 euros in total for working a 5-day week.

In his new job, Pierre is paid 14 euros per hour.

In his new job, Pierre works for 7 hours each day for a 5-day week.

- (a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week.

..... %

Marie changes her job.

Her salary decreases by 6%

Her new salary is 23 030 euros.

- (b) Work out Marie's salary before she changes her job.

..... euros

(Total for Question 6 is 7 marks)

Q20 **PAST PAPER** Edexcel 4MA1 • May 2023 • Paper 1H • Q3**CH 1 • TOPIC 6** **4 marks**

- 3 Nanette buys 60 notebooks for a total cost of 780 dirhams.

Nanette sells 70% of the notebooks for 22 dirhams each.
She sells the remaining notebooks for 19 dirhams each.

Work out Nanette's percentage profit.

Give your answer correct to 3 significant figures.

..... %

(Total for Question 3 is 4 marks)

Q21**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q3

CH 1 • TOPIC 6 4 marks

3 Joshua buys a car for \$12 500

He sells the car to Nina.

Nina pays

- a deposit of \$1500
- followed by 36 monthly payments of \$450

Work out Joshua's percentage profit.

..... %

(Total for Question 3 is 4 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q2

CH 1 • TOPIC 6 6 marks

2 Anna makes cups.

Each cup costs 6 Swiss francs to make.

Anna puts the cups into boxes to sell.

Each box contains 4 cups.

Anna sells 80 boxes of cups for a total of 2160 Swiss francs.

(a) Work out the percentage profit Anna makes.

Show your working clearly.

..... %

The height of each cup is 9 cm, correct to the nearest cm

(b) Write down the lower bound of the height.

..... cm

The weight of each cup is 120 g, correct to the nearest 10 g

(c) Write down the upper bound of the weight.

..... g

(Total for Question 2 is 6 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q4

CH 1 • TOPIC 6**5 marks**

- 4 Joshua is going to cover a floor with tiles for a customer.
The area of the floor is 45 m^2

Joshua buys one box of tiles for each 1.5 m^2 of floor area.
Each box of tiles costs £64

Joshua also buys 5 bags of tile adhesive.
Each bag of tile adhesive costs £12

Joshua charges the customer £3000

Work out his percentage profit.
Give your answer correct to one decimal place.

..... %

(Total for Question 4 is 5 marks)

03 Reverse Percentages

You are given the value **after** the change. Write the multiplier, then **divide**. Never take the percentage of the new price.

WHEN YOU SEE → YOU DO

“sale price ... reduced by 15% ... find the normal price”

→ sale $\div$ 0.85

“943 is 15% more than March”

→ $943 \div 1.15$

“17% of N is 357”

→ $357 \div 0.17$

“5.48 million was 22% of the population”

→ $5.48 \div 0.22$

“by how much was the price reduced?”

→ original – sale

Multiplier

New $\div$ multiplier

Check forwards

WORKED MODEL • JANUARY 2020 P1HR

A hat is reduced by 15%. The sale price is 20.40 euros. Find the normal price.

1 multiplier $1 - 0.15 = 0.85$

write the multiplier

2 $20.40 \div 0.85$

backwards: divide

ANSWER 24 euros

× **TRAP** $20.40 \times 1.15 = 23.46$ is wrong. 15% of the sale price is not 15% of the normal price.

CALCULATOR CHECK Check forwards: $24 \times 0.85 = 20.40$ — correct

IDEA 03

Practise: Reverse Percentages

My questions first, then selected practice, then real past papers.

Q24 DR ESLAM Backwards

CH 1 • TOPIC 6 3 marks

After a 12% increase, a train ticket costs £39.20. What did it cost before?

SELECTED PRACTICE**Q25** PRACTICE Practice • Easy

CH 1 • TOPIC 6 3 marks

14 Ed has a card shop.

Ed's profit on "Good Luck" cards in 2018 was £360. This was a decrease of 20% on his profit in 2017.

Work out Ed's profit on "Good Luck" cards in 2017.

£

(3 marks)

Q26**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6 5 marks

- 12 (a)** Each year Wenford Hospital records how long patients wait to be treated in the Accident and Emergency department.

In 2015 patients waited 11% less time than in 2014

In 2015 the average time patients waited was 68 minutes.

Work out the average time patients waited in 2014

Give your answer to the nearest minute.

(3 marks)

- (b)** The hospital has a target to reduce the average time patients wait to be treated in the Accident and Emergency department to 60 minutes in 2016.

Work out the percentage decrease from 68 minutes to 60 minutes.

(2 marks)

Q27**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6 6 marks

- 17 (a)** Lijuan's salary is 180000 Hong Kong Dollars (HK\$).
She gets a salary increase of 3%

Work out Lijuan's salary after this increase.

(3 marks)

- (b)** In a sale, all normal prices are reduced by 15%
The sale price of a camera is HK\$6630

Work out the normal price of the camera.

(3 marks)

Q28**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6

4 marks

4 A shop has a sale.

Microwave ovens

$\frac{1}{3}$ off normal price

Combination ovens

40% off normal price

A microwave oven has a sale price of £90

A combination oven has a sale price of £84

Which of these ovens has the greater normal price?

You must show all your working.

(4 marks)

PAST PAPERS

Q29**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q7

CH 1 • TOPIC 6 3 marks

- 7 In a sale, the normal price of a hat is reduced by 15%
The sale price of the hat is 20.40 euros.

Work out the normal price of the hat.

..... euros

(Total for Question 7 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q9

CH 1 • TOPIC 6 3 marks

- 9 In a sale, normal prices are reduced by 20%
A bag costs 1080 rupees in the sale.

Work out the normal price of the bag.

..... rupees

(Total for Question 9 is 3 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q6

CH 1 • TOPIC 6**3 marks**

- 6 On Wednesday, the price of 1 litre of petrol was £1.26
The price of petrol on Wednesday was 5% more than the price of petrol on the previous Monday.

Calculate the price of 30 litres of petrol on the previous Monday.

£.....

(Total for Question 6 is 3 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q8

CH 1 • TOPIC 6**3 marks**

- 8 In 2018, the population of Sydney was 5.48 million.
This was 22% of the total population of Australia.

Work out the total population of Australia in 2018
Give your answer correct to 3 significant figures.

..... million

(Total for Question 8 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q8

CH 1 • TOPIC 6 3 marks

- 8 A cinema increased the cost of an adult ticket by 12%

After the increase, the cost of an adult ticket was £18.20

Work out the cost of an adult ticket before the increase.

£.....

(Total for Question 8 is 3 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • November 2023 • Paper 1H • Q6**CH 1 • TOPIC 6** **4 marks**

- 6 Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

ticket A $\frac{1}{6}$ off normal price

ticket B 20% off normal price

The sale price of ticket **A** is 140 euros.

The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

..... euros

(Total for Question 6 is 4 marks)

Q35**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q8

CH 1 • TOPIC 6 3 marks

- 8 In a sale, all normal prices are reduced by 17%

The sale price of a fridge is 6225 rupees.

Work out the normal price of the fridge.

..... rupees

(Total for Question 8 is 3 marks)

Q36**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q4

CH 1 • TOPIC 6 3 marks

- 4 Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.

This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

..... kilometres

(Total for Question 4 is 3 marks)

7 Shop A and Shop B have offers for buying the same type of laptop.

The normal price of the laptop in Shop A is different to the normal price of the laptop in Shop B

Shop A	Shop B
Our normal price £475	Get 15% off our normal price
Get 16% off our normal price	Only pay £408

Which shop gives more money off their normal price?
Show your working clearly.

(Total for Question 7 is 4 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q4

CH 1 • TOPIC 6 5 marks

- 4 N is a number.
17% of N is 357

(a) Work out the value of N

$N =$

In 2019, the population of a village was 650
In 2020, the population of the village was 806

- (b) Work out the percentage increase in the population.

..... %

(Total for Question 4 is 5 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q5

CH 1 • TOPIC 6 3 marks

- 5 In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.

Work out the normal price of the watch.

..... euros

(Total for Question 5 is 3 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q9

CH 1 • TOPIC 6 3 marks

- 9 In a sale, normal prices are reduced by 15%

The sale price of a dishwasher is 612 Swiss francs.

Work out the normal price of the dishwasher.

..... Swiss francs

(Total for Question 9 is 3 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q9

CH 1 • TOPIC 6 3 marks

- 9 Hari scored 140 points playing a game on Tuesday.

This was 12% more points than he scored playing the game on Monday.

Work out the number of points that Hari scored playing the game on Monday.

.....

(Total for Question 9 is 3 marks)

04 Repeated and Combined Changes

Two changes one after the other: multiply the multipliers. Percentages do **not** add.

WHEN YOU SEE → YOU DO

“increased by 15% and then decreased by 8%”

→ $1.15 \times 0.92 = 1.058$: up 5.8%

“4 + 2.5 = 6.5 ... is he right?”

→ $0.96 \times 0.975 = 0.936$: 6.4%, not 6.5%

“reduced by 20%, then by 30%”

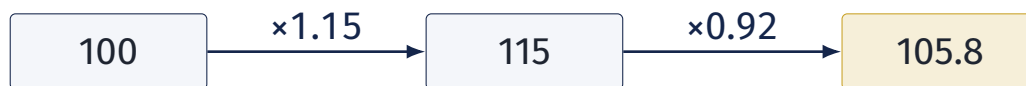
→ $0.8 \times 0.7 = 0.56$: 44% off, not 50%

“2% less every hour”, $V_{t+1} = kV_t$

→ $k = 0.98$

length -15%, width -10%

→ work out the new perimeter / area, then the %



overall multiplier $1.15 \times 0.92 = 1.058 \rightarrow$ up 5.8%, not 7%

× **TRAP** Adding or subtracting percentages ($15 - 8 = 7\%$). Each change acts on a different amount.

IDEA 04

Practise: Repeated and Combined Changes

My questions first, then selected practice.

Q42 DR ESLAM Twice

CH 1 • TOPIC 6 2 marks

A price rises by 10% and then falls by 10%. Find the overall percentage change.

SELECTED PRACTICE

Q43

PRACTICE

Practice • Easy

CH 1 • TOPIC 6 3 marks

12 The table shows information about the population of a city.

Population in 2001	Population in 2011
420 000	480 000

Liam claims,

“From 2011 to 2021 the population of the city will increase by the same percentage as from 2001 to 2011”

He works out,

$$\text{population increase from 2001 to 2011} = 480\,000 - 420\,000$$

$$= 60\,000$$

$$\text{population in 2021} = 480\,000 + 60\,000$$

$$= 540\,000$$

Does the population of 540 000 match his claim? You **must** show your working.

(3 marks)

Q44**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6**2 marks**

15 In a shop, all normal prices are reduced by 20% to give the sale price.

The sale price of a TV set is then reduced by 30%.

Mary says,

" $30 + 20 = 50$, so this means that the normal price of the TV set has been reduced by 50% "

Is Mary right?

You must give a reason for your answer.

(2 marks)

Q45**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6**5 marks**

2 (a) Ibrar bought a house for £145 000

The value of the house depreciated by 4% in the first year. The value of the house depreciated by 2.5% in the second year.

Ibrar says,

" $4 + 2.5 = 6.5$ so in two years the value of my house depreciated by 6.5%"

Is Ibrar right?

You must give a reason for your answer.

(2 marks)

(b) The value of Ibrar's house increases by $x\%$ in the third year.

At the end of the third year the value of Ibrar's house is £140 000.

Work out the value of x .

Give your answer correct to 3 significant figures.

(3 marks)

Q46**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6 2 marks

- 9 During last season the cost of a ticket to watch Seapron United increased by 15% and then decreased by 8%.

Work out the overall percentage change in the cost of a ticket to watch Seapron United during last season.

(2 marks)**Q47****PRACTICE**

Practice • Hard

CH 1 • TOPIC 6 2 marks

- 12 (a) Show that $a\%$ of $b = b\%$ of a

(1 mark)

- (b) Rosie says,

"160% of 40 = 140% of 60 because $a\%$ of $b = b\%$ of a "

Is she correct?

Tick a box.

☐

Yes

☐

No

Give a reason for your answer.

(1 mark)

Q48**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6**6 marks**

- 18** In 2017, the value of a house increased by 4%.
In 2018, the value of the house then decreased by 3%.

Teresa says

Over the two years the value of the house increased by exactly 1% because $4 - 3 = 1$.

Show that Teresa is wrong.

(6 marks)

05 Percentages with Fractions and Ratio

Mixed questions: split the whole with the ratio or the fractions first, then take each percentage of its own part.

WHEN YOU SEE → YOU DO

“ratio 9 : 4 : 2 ... 45% of the tulips ...”

→ share the total first, then the % of each share

“ $\frac{3}{8}$... 45% ... and \$406”

→ the \$406 is what is left: $1 - \frac{3}{8} - 0.45$

“45% more than Abel”

→ Abel = a , Bahira = $1.45a$

“what fraction ... give in its simplest form”

→ add the parts, cancel

WORKED MODEL • JUNE 2024 P2H

600 bulbs, tulip : crocus : daffodil = 9 : 4 : 2. Yellow: 45% of tulips, $\frac{5}{8}$ of crocus, all daffodils.

1 $600 \div 15 = 40$: 360, 160, 80

share by the ratio

2 $0.45 \times 360 = 162$, $\frac{5}{8} \times 160 = 100$, 80

each part

ANSWER $162 + 100 + 80 = 342$ yellow

IDEA 05

Practise: Percentages with Fractions and Ratio

My questions first, then selected practice, then real past papers.

Q49**DR ESLAM**

Split first

CH 1 • TOPIC 6 3 marks

80 students: boys : girls = 3 : 5. 40% of the boys and 60% of the girls walk to school. How many walk?

SELECTED PRACTICE**Q50****PRACTICE**

Practice • Hard

CH 1 • TOPIC 6 3 marks

- 6 Marie invests £8000 in an account for one year.
At the end of the year, interest is added to her account.

Marie pays tax on this interest at a rate of 20%
She pays £28.80 tax.

Work out the percentage interest rate for the account.

(3 marks)

Q51**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6**4 marks**

13 When you earn money you pay income tax.

The amount you pay depends on how much you earn that year. You pay

0% on the first £12500 you earn

20% on the next £37 500 you earn

40% on the next £112 500 you earn.

One year, Kim paid £9260 income tax.

Work out how much she earned that year.

£.....

(4 marks)

Q52

PRACTICE

Practice • Hard

CH 1 • TOPIC 66 marks

PRACTISE - 1.6

15 Lily buys and sells microwaves.

She buys each one for £32 and sells it for £60.
She also pays £7 for the delivery of each microwave she sells.

If she sells a microwave that is faulty then Lily must pay for its repair and redelivery.
This costs her another £25 for each faulty microwave.

Last month, 6 out of the 80 microwaves Lily sold were faulty.

This month she has orders for 133 microwaves.
Calculate her expected percentage profit on this month’s order. Showing your working
in the boxes below may help you present your work.

Expected number of faulty microwaves:	Expected costs:
Income from sales:	Expected percentage profit:

.....%

(6 marks)

Q53**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6 6 marks

19 In a village the ratio of males to females is 2:1.

40% of the people in the village are right-handed males.

25% of the people in the village are right-handed females.

Show that the proportion of females who are right-handed is greater than the proportion of males who are right-handed.

(6 marks)

PAST PAPERS

Q54 **PAST PAPER** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q10**CH 1 • TOPIC 6** **5 marks****10** Alex makes 80 cakes to sell.

He makes chocolate cakes, lemon cakes and fruit cakes where

$$\begin{array}{ccccc} \text{number of} & & \text{number of} & & \text{number of} \\ \text{chocolate cakes} & : & \text{lemon cakes} & : & \text{fruit cakes} \end{array} = 3 : 2 : 5$$

Alex sells

all of the chocolate cakes

 $\frac{3}{4}$ of the lemon cakes $\frac{7}{8}$ of the fruit cakes

The profit he makes on each cake he sells is shown in the table.

Type of cake	Profit per cake he sells
chocolate	£2.00
lemon	£1.70
fruit	£2.40

Work out the total profit that Alex makes from the cakes he sells.

£.....

(Total for Question 10 is 5 marks)

Q55**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q10

CH 1 • TOPIC 6 3 marks

10 The people working for a company work in Team A or in Team B.

number of people in Team A : number of people in Team B = 3 : 4

$\frac{4}{5}$ of Team A work full time.

24% of Team B work full time.

Work out what fraction of the people working for the company work full time.
Give your fraction in its simplest form.

.....
(Total for Question 10 is 3 marks)

Q56**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q6

CH 1 • TOPIC 6 4 marks

- 6 Mariana sells bags of bird food.

The bags that Mariana sold last week each contained 12 kg of seeds.

The bags that she is going to sell next week will each contain a mixture of nuts and seeds where for each bag

$$\text{weight of nuts : weight of seeds} = 4 : 5$$

The total weight of the nuts and the seeds in each bag will be 19.35 kg

The weight of seeds in each bag that Mariana sells next week will be less than the weight of seeds in each bag that Mariana sold last week.

Work out this decrease as a percentage of the weight of seeds in each bag that Mariana sold last week.

Give your answer correct to one decimal place.

..... %

(Total for Question 6 is 4 marks)

Q57**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q4

CH 1 • TOPIC 6**4 marks**

- 4 The language department of a college has 180 students.
Each student studies exactly one of French, German, Italian or Spanish.

15 students study French.

45% of the students study German.

Express the percentage of students studying Italian or Spanish as a percentage of those studying French or German.

.....%

(Total for Question 4 is 4 marks)

Q58**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q13

CH 1 • TOPIC 6 5 marks**13** Antoine is going on holiday.

He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Antoine pays to the holiday company.

Payment	Amount paid
Payment 1	$\frac{3}{8}$ of the total cost
Payment 2	45% of the total cost
Payment 3	\$406

Work out how much Antoine has to pay for Payment 2

\$

(Total for Question 13 is 5 marks)

Q59**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q8

CH 1 • TOPIC 6 5 marks

8 Behnaz makes 300 celebration cards so that

number of birthday cards : number of anniversary cards : number of congratulations cards = 7:5:3

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.

None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.

Give your answer in its simplest form.

.....
(Total for Question 8 is 5 marks)

Q60**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q8

CH 1 • TOPIC 6 5 marks

- 8 Ishir plants 600 bulbs in a garden.
He plants tulip bulbs, crocus bulbs and daffodil bulbs so that

number of tulip bulbs : number of crocus bulbs : number of daffodil bulbs = 9 : 4 : 2

45% of the tulip bulbs are for yellow flowers.

$\frac{5}{8}$ of the crocus bulbs are for yellow flowers.

All of the daffodil bulbs are for yellow flowers.

Work out the number of bulbs that are for yellow flowers.

(Total for Question 8 is 5 marks)

Q61**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q14

CH 1 • TOPIC 6**5 marks**

14 The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are $\frac{3}{2}$ times the savings of Chanda.

Work out the savings of Chanda.

..... dinars

(Total for Question 14 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE UP

$$\times(1 + r)$$

ZONE DOWN

$$\times(1 - r)$$

ZONE FORWARDS

$$\times m$$

ZONE BACKWARDS

$$\div m$$

- | | | |
|-----------|------------------------|--|
| 01 | Of an amount | Decimal $\times$ amount; VAT $\times 1.2$. |
| 02 | Change / profit | $\frac{\text{change}}{\text{original}} \times 100$ — original on the bottom. |
| 03 | Reverse | New $\div$ multiplier. Never a % of the new value. |
| 04 | Repeated | Multiply the multipliers. Percentages do not add. |
| 05 | Mixed | Split the whole first, then each %. |

BOTTOM LINE Multiplier first $\rightarrow$ forwards $\times \rightarrow$ backwards $\div \rightarrow$ twice: multiply them.

Answers

Percentages · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • PERCENTAGE OF AN AMOUNT

Q1	Dr Eslam · Multiplier	$85 \times 1.2 = \text{£}102$
Q2	Practice · Easy	$\text{£}900$
Q3	Practice · Easy	400%
Q4	Practice · Medium	$650 \text{ L: } \text{£}414.96$
Q5	Practice · Medium	$20\,800 + 4\,600 = \text{£}25\,400$: Yes
Q6	Edexcel 4MA1 · May 2024 · Paper 1HR · Q3	$120 \times 0.60 + 80 \times 1.05 = \text{£}156$

IDEA 02 • PERCENTAGE CHANGE AND PROFIT

Q7	Dr Eslam • Percentage change	$\frac{2700}{12500} \times 100 = 21.6\%$
Q8	Practice • Easy	8.49%
Q9	Practice • Easy	$552 - 425 = 127$: 23.0%
Q10	Practice • Easy	75.3%
Q11	Practice • Medium	30%
Q12	Edexcel 4MA1 • January 2020 • Paper 2H • Q8	(a) 9% then 8.6%: No (b) 168 euros
Q13	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q5	(a) 10.2% (b) 45 euros
Q14	Edexcel 4MA1 • January 2021 • Paper 2H • Q3	(a) 11.8% (b) \$484
Q15	Edexcel 4MA1 • November 2021 • Paper 2H • Q6	40%
Q16	Edexcel 4MA1 • January 2022 • Paper 1H • Q4	income £1644: 44%
Q17	Edexcel 4MA1 • January 2022 • Paper 2H • Q8	(a) 1.5% (b) £215 000
Q18	Edexcel 4MA1 • January 2022 • Paper 2HR • Q2	(a) $P = 30$ (b) A saves \$288, B saves \$300: B
Q19	Edexcel 4MA1 • June 2023 • Paper 2HR • Q6	(a) 22.5% (b) 24 500 euros
Q20	Edexcel 4MA1 • May 2023 • Paper 1H • Q3	62.3%
Q21	Edexcel 4MA1 • November 2023 • Paper 1H • Q3	41.6%
Q22	Edexcel 4MA1 • June 2025 • Paper 1H • Q2	(a) 12.5% (b) 8.5 cm (c) 125 g
Q23	Edexcel 4MA1 • June 2025 • Paper 2HR • Q4	cost £1980: 51.5%

IDEA 03 • REVERSE PERCENTAGES

		39.20 ÷ 1.12 = £35
Q24	Dr Eslam • Backwards	
		360 ÷ 0.8 = £450
Q25	Practice • Easy	
		(a) 76 minutes (b) 11.8%
Q26	Practice • Medium	
		(a) HK\$185 400 (b) HK\$7800
Q27	Practice • Medium	
		microwave £135, combination £140: the
Q28	Practice • Hard	combination oven
		24 euros
Q29	Edexcel 4MA1 • January 2020 • Paper 1HR • Q7	
		1350 rupees
Q30	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q9	
		£36
Q31	Edexcel 4MA1 • January 2021 • Paper 2HR • Q6	
		24.9 million
Q32	Edexcel 4MA1 • November 2021 • Paper 1H • Q8	
		£16.25
Q33	Edexcel 4MA1 • June 2022 • Pa- per 2HR • Q8	
		A 168, B 170: 2 euros
Q34	Edexcel 4MA1 • November 2023 • Paper 1H • Q6	
		7500 rupees
Q35	Edexcel 4MA1 • June 2024 • Pa- per 2HR • Q8	
		820 km
Q36	Edexcel 4MA1 • May 2024 • Pa- per 1H • Q4	
		A £76 off, B £72 off: Shop A
Q37	Edexcel 4MA1 • November 2024 • Paper 1H • Q7	
		(a) 2100 (b) 24%
Q38	Edexcel 4MA1 • November 2024 • Paper 2H • Q4	
		275 euros
Q39	Edexcel 4MA1 • June 2025 • Pa- per 1H • Q5	
		720 Swiss francs
Q40	Edexcel 4MA1 • June 2025 • Pa- per 2HR • Q9	

IDEA 04 • REPEATED AND COMBINED CHANGES**Q42** Dr Eslam • Twice

$$1.1 \times 0.9 = 0.99: 1\% \text{ decrease}$$

Q43 Practice • Easy

+14.3% gives about 548 571, not 540 000: No

Q44 Practice • Medium

No: $0.8 \times 0.7 = 0.56$, a 44% reduction

Q45 Practice • Hard

(a) No: $0.96 \times 0.975 = 0.936$, 6.4% (b) $x = 3.15$

Q46 Practice • Hard

5.8% increase

Q47 Practice • Hard

(a) $\frac{a}{100}b = \frac{b}{100}a$ (b) No: $64 \neq 84$

Q48 Practice • Hard

$$1.04 \times 0.97 = 1.0088: \text{up } 0.88\%, \text{ not } 1\%$$

IDEA 05 • PERCENTAGES WITH FRACTIONS AND RATIO

- Q49** Dr Eslam • Split first 30 boys, 50 girls: $12 + 30 = 42$
- Q50** Practice • Hard interest £144: 1.8%
- Q51** Practice • Hard £54 400
- Q52** Practice • Hard cost £5436.38, income £7980: 46.8%
- Q53** Practice • Hard females $\frac{0.25}{1/3} = 75\% > \text{males } \frac{0.40}{2/3} = 60\%$
- Q54** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q10 $48 + 20.40 + 84 = £152.40$
- Q55** Edexcel 4MA1 • January 2021 • Paper 2H • Q10 $\frac{12}{25}$
- Q56** Edexcel 4MA1 • May 2021 • Paper 2H • Q6 10.4%
- Q57** Edexcel 4MA1 • June 2022 • Paper 1HR • Q4 $\frac{84}{96} = 87.5\%$
- Q58** Edexcel 4MA1 • June 2022 • Paper 1H • Q13 total \$2320: \$1044
- Q59** Edexcel 4MA1 • June 2022 • Paper 2H • Q8 $\frac{23}{75}$
- Q60** Edexcel 4MA1 • June 2024 • Paper 2H • Q8 342
- Q61** Edexcel 4MA1 • May 2024 • Paper 1H • Q14 Abel 6300, Bahira 9135: Chanda 6090 dinars